Quantum-Inspired Multi-Omics Fusion Framework for Interpretable Disease Classification

Abstract

The integration of heterogeneous multi-omics data has emerged as a critical challenge in computational biology and precision medicine. While individual omics modalities such as genomics, transcriptomics, and proteomics provide valuable biological insights, effective multimodal fusion remains difficult due to high dimensionality, modality imbalance, redundancy, and complex nonlinear relationships. In this study, we propose a quantum-inspired multi-omics fusion framework incorporating attention-guided modality gating and interpretable deep learning for robust disease classification.

The proposed architecture integrates genomics, transcriptomics, and proteomics representations using a gated fusion mechanism capable of dynamically weighting modality contributions. The framework was evaluated on preprocessed multi-omics datasets using stratified 5-fold cross-validation. Model performance was assessed using ROC-AUC, F1-score, precision-recall analysis, calibration curves, confusion matrices, permutation importance, and SHAP explainability.

Experimental results demonstrated strong predictive performance with a mean ROC-AUC of approximately 0.972 and robust cross-validation stability. Transcriptomics emerged as the dominant predictive modality through attention-gate analysis, while genomics provided complementary contextual information that enhanced multimodal integration performance. Comparative experiments against classical machine learning baselines including Logistic Regression, Support Vector Machines (SVM), Random Forest, XGBoost, and Multilayer Perceptrons (MLP) showed that the proposed fusion architecture outperformed conventional methods in multimodal integration tasks.

Interpretability analyses revealed sparse but biologically meaningful feature contributions, supporting the framework's ability to identify influential biomarkers and modality interactions. The proposed system demonstrates the potential of quantum-inspired multimodal fusion architectures for interpretable and high-performance computational disease modeling.

Keywords: Multi-omics integration, Quantum-inspired learning, Deep learning, Bioinformatics, Explainable AI, SHAP, Precision medicine, Multimodal fusion

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1. Introduction

1.1 Background

Advances in high-throughput sequencing technologies have enabled the generation of large-scale biological datasets spanning multiple molecular layers, including genomics, transcriptomics, proteomics, epigenomics, and metabolomics. These multi-omics datasets provide complementary perspectives on biological systems and disease mechanisms.

Genomics captures inherited and acquired DNA-level variations, transcriptomics reflects gene expression dynamics, and proteomics represents downstream functional molecular activity. Together, these modalities provide a comprehensive understanding of cellular states and disease progression.

However, effective integration of heterogeneous multi-omics data remains a major challenge in computational biology due to:

- High dimensionality
- Noise and sparsity
- Modality imbalance
- Nonlinear cross-modal relationships
- Limited sample sizes
- Missing modality interactions

Traditional machine learning approaches often rely on direct feature concatenation, which can introduce redundancy and reduce predictive performance. Deep learning-based fusion architectures offer improved representational learning capabilities but frequently suffer from poor interpretability and unstable multimodal integration.

Recent developments in quantum-inspired machine learning have introduced new perspectives for modeling high-dimensional complex systems through enhanced representation learning and nonlinear information encoding.

1.2 Motivation

Most existing multi-omics integration systems encounter one or more of the following limitations:

1. Poor handling of heterogeneous modalities
2. Lack of interpretability
3. Overfitting in high-dimensional settings
4. Inefficient modality weighting
5. Weak biological consistency
6. Limited robustness across folds and datasets

To address these challenges, this study proposes a quantum-inspired gated fusion architecture capable of:

- Dynamically learning modality importance
- Integrating complementary biological information
- Providing interpretable feature attribution
- Improving classification robustness
- Enhancing multimodal representation learning

1.3 Objectives

The major objectives of this work are:

1. To design a quantum-inspired multi-omics fusion framework for disease classification.
2. To integrate genomics, transcriptomics, and proteomics data using attention-guided gating.
3. To evaluate multimodal learning performance using robust cross-validation.
4. To compare the proposed system against classical machine learning baselines.
5. To investigate modality importance using attention weights and permutation importance.
6. To improve model interpretability using SHAP explainability.

1.4 Contributions

The major contributions of this study include:

- Development of a quantum-inspired multi-omics fusion architecture.
- Implementation of dynamic attention-based modality gating.
- Comprehensive evaluation using ROC, PR, calibration, and cross-validation analyses.
- Integration of SHAP-based explainability and permutation importance.
- Comparative benchmarking against multiple classical machine learning models.
- Biological interpretation of modality interactions and feature importance.

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2. Literature Review

2.1 Multi-Omics Integration

Multi-omics integration has become increasingly important in precision medicine and systems biology. Traditional approaches include:

- Early feature concatenation
- Matrix factorization
- Graph-based integration
- Bayesian integration
- Kernel learning methods

Although these approaches provide useful insights, many fail to effectively capture nonlinear interactions among modalities.

2.2 Deep Learning in Bioinformatics

Deep learning models such as autoencoders, CNNs, GNNs, and transformers have been applied to biological data integration. These models improve nonlinear representation learning but often require large datasets and may lack interpretability.

Recent studies have shown that attention mechanisms improve multimodal learning by selectively emphasizing important modalities and features.

2.3 Explainable AI in Biomedical Research

Interpretability is essential in biomedical machine learning due to the need for clinical trust and biological validation.

Common explainability approaches include:

- SHAP (SHapley Additive exPlanations)
- LIME
- Attention visualization
- Permutation importance
- Integrated gradients

SHAP has become particularly popular because of its theoretical grounding and ability to provide local and global feature explanations.

2.4 Quantum-Inspired Learning

Quantum-inspired machine learning incorporates principles such as:

- High-dimensional state representations
- Superposition-inspired feature encoding
- Enhanced nonlinear transformation
- Probabilistic representation learning

Although true quantum computing remains computationally expensive, quantum-inspired architectures can improve representation capacity in classical deep learning systems.

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3. Materials and Methods

3.1 Dataset

The study utilized preprocessed multi-omics datasets containing:

- Genomics features
- Transcriptomics features
- Proteomics features

After preprocessing and dimensionality reduction, each modality contained 64 features.

Final dataset dimensions:

Modality	Shape
Genomics	(626, 64)
Transcriptomics	(626, 64)
Proteomics	(626, 64)

The dataset consisted of binary disease classification labels.

3.2 Data Preprocessing

The preprocessing pipeline included:

1. Sample alignment
2. Missing value handling
3. Feature normalization
4. Dimensionality reduction
5. Label encoding
6. Dataset balancing

All modalities were standardized before model training.

3.3 Proposed Architecture

3.3.1 Overview

The proposed architecture consists of:

1. Modality-specific encoders
2. Quantum-inspired latent representation blocks
3. Attention-guided modality gates
4. Fusion layer
5. Classification head

3.3.2 Modality Encoders

Each omics modality was processed independently using dedicated neural encoders to learn latent feature representations.

3.3.3 Quantum-Inspired Representation Layer

Quantum-inspired transformations were introduced to enhance nonlinear feature interactions and improve latent representation capacity.

3.3.4 Attention-Guided Gating

A dynamic gating mechanism learned modality importance during training.

The gate weights were computed adaptively for each sample, enabling the model to emphasize informative modalities while suppressing noisy signals.

3.3.5 Fusion Module

The gated latent representations were fused into a unified multimodal embedding for downstream classification.

3.4 Training Strategy

The model was trained using:

- Stratified 5-fold cross-validation
- Adam optimizer
- Early stopping
- Weighted loss optimization
- Learning rate scheduling

3.5 Evaluation Metrics

Performance was evaluated using:

- ROC-AUC
- F1-score
- Precision
- Recall
- Average Precision (PR-AUC)
- Calibration curves
- Confusion matrices

3.6 Explainability Methods

3.6.1 SHAP Analysis

SHAP values were computed to estimate feature contributions and identify influential biomarkers.

3.6.2 Permutation Importance

Permutation importance was used to estimate modality-level importance by measuring performance degradation after feature shuffling.

3.6.3 Attention Gate Analysis

The learned gate weights were analyzed to understand modality contribution patterns.

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4. Experimental Results

4.1 Cross-Validation Performance

The proposed framework demonstrated stable performance across folds.

Metric	Mean ± SD
ROC-AUC	0.944 ± 0.026
F1-score	0.915 ± 0.032

The low variance across folds indicates robust generalization and reduced overfitting.

4.2 ROC and Precision-Recall Analysis

The model achieved:

Metric	Value
ROC-AUC	0.972
Average Precision	0.992

The high ROC-AUC indicates strong class separability, while the high average precision demonstrates robust positive prediction performance.

4.3 Confusion Matrix Analysis

The confusion matrix results showed:

Prediction Outcome	Count
True Positives	69
True Negatives	21
False Positives	2
False Negatives	2

The model achieved low false-negative and false-positive rates, which is particularly important in biomedical applications.

4.4 Attention Gate Analysis

The learned modality gate weights were:

Modality	Mean Gate Weight
Transcriptomics	0.768
Proteomics	0.203
Genomics	0.028

Interpretation

Transcriptomics emerged as the dominant predictive modality, suggesting that dynamic gene expression states provide the strongest direct disease-discriminative signal.

Proteomics contributed secondary functional information, while genomics provided limited direct predictive importance.

4.5 Permutation Importance

Permutation importance analysis produced the following AUC drops:

Modality	AUC Drop
Genomics	0.1635
Transcriptomics	0.0563
Proteomics	0.0266

Interpretation

Interestingly, genomics caused the largest performance degradation when shuffled despite receiving the lowest attention weight.

This suggests that genomics contributes important contextual and structural information that supports multimodal integration.

The findings indicate that:

- Transcriptomics provides dominant direct predictive signal
- Genomics contributes complementary contextual information
- Proteomics refines functional representations

4.6 Calibration Analysis

The calibration curve demonstrated that predicted probabilities closely followed the ideal calibration diagonal.

This indicates:

- Reliable confidence estimation
- Reduced overconfidence
- Better clinical interpretability

4.7 SHAP Explainability Analysis

SHAP analysis revealed sparse but biologically meaningful feature contributions.

Key observations include:

- Most features contributed minimally
- A subset of features strongly influenced predictions
- Both positive and negative feature effects were observed

These results suggest that the model successfully identified discriminative molecular patterns rather than relying on random noise.

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5. Baseline Comparison

5.1 Classical Machine Learning Models

The proposed framework was compared against several classical machine learning baselines:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest
- XGBoost
- Multilayer Perceptron (MLP)

5.2 Baseline Results

Dataset	Best Baseline Model	AUC	F1
Transcriptomics	SVM	0.9532	0.9257
T+P	SVM	0.9529	0.9197
All Modalities	SVM	0.9502	0.9170
G+T	SVM	0.9475	0.9001
Proteomics	Random Forest	0.9422	0.8940
G+P	SVM	0.9403	0.8898

Dataset	Best Baseline Model	AUC	F1
Genomics	XGBoost	0.9049	0.8913

5.3 Comparison with Proposed Model

Model	AUC	F1
Best Classical Baseline	0.9532	0.9257
Proposed Fusion Model	0.972	0.915

5.4 Interpretation

The proposed multimodal fusion framework outperformed all classical baselines in ROC-AUC.

An important observation was that naive feature concatenation often degraded classical machine learning performance, indicating difficulty in integrating heterogeneous modalities.

In contrast, the proposed gated fusion architecture successfully leveraged complementary multimodal information.

These findings demonstrate that:

1. Transcriptomics alone provides strong predictive power.
2. Classical models struggle with multimodal heterogeneity.
3. The proposed architecture effectively integrates complementary biological information.
4. Attention-guided fusion improves multimodal learning.

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6. Discussion

The experimental results demonstrate that the proposed quantum-inspired multimodal fusion framework effectively integrates heterogeneous biological modalities for disease classification.

6.1 Biological Interpretation

The dominance of transcriptomics suggests that dynamic cellular states are more informative for classification than static genomic variation.

This aligns with biological understanding that:

- Gene expression reflects active disease mechanisms
- Proteins represent downstream functional behavior
- Genomic mutations provide contextual predisposition

The discrepancy between attention weights and permutation importance indicates sophisticated modality interaction behavior.

Although genomics received low direct attention weights, its large permutation impact suggests that it contributes structural latent information supporting transcriptomic interpretation.

6.2 Multimodal Fusion Behavior

The baseline experiments revealed that simple feature concatenation often reduced classical machine learning performance.

This highlights the challenge of integrating heterogeneous omics modalities.

The proposed gated fusion architecture overcame this limitation by:

- Dynamically weighting modalities
- Suppressing noisy signals
- Enhancing complementary information
- Learning nonlinear cross-modal interactions

6.3 Explainability and Clinical Relevance

The integration of SHAP explainability, permutation importance, and attention analysis improved model transparency.

Interpretability is essential for:

- Biomarker discovery
- Clinical trust
- Biological validation
- Precision medicine applications

The sparse SHAP patterns suggest that the model identified meaningful molecular signatures rather than overfitting to noise.

6.4 Limitations

Despite strong performance, the study has several limitations:

1. Limited sample size
2. Single dataset evaluation
3. Lack of external validation
4. Reduced biological annotation of SHAP features
5. Absence of pathway enrichment analysis

Future work should address these limitations through:

- Larger datasets
- Independent cohort validation
- Multi-center evaluation
- Biological pathway analysis
- Integration of additional omics modalities

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7. Conclusion

This study presented a quantum-inspired multi-omics fusion framework for interpretable disease classification.

The proposed architecture successfully integrated genomics, transcriptomics, and proteomics data using attention-guided gating and nonlinear multimodal fusion.

Experimental evaluation demonstrated:

- Strong classification performance
- Stable cross-validation behavior
- Improved multimodal integration
- Robust explainability
- Biological interpretability

The framework achieved superior ROC-AUC performance compared to classical machine learning baselines and revealed meaningful modality interaction patterns.

Transcriptomics emerged as the dominant direct predictive modality, while genomics contributed complementary contextual information that improved multimodal learning.

The results demonstrate the potential of quantum-inspired multimodal fusion architectures for computational biology and precision medicine applications.

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8. Future Work

Future research directions include:

1. External dataset validation
2. Pathway enrichment analysis
3. Real quantum circuit integration
4. Transformer-based multimodal fusion
5. Graph neural network integration
6. Survival prediction modeling

7. Single-cell multi-omics integration
8. Clinical deployment optimization

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9. Appendix

Appendix A: Hyperparameter Configuration

The proposed framework was trained using the following hyperparameters:

Parameter	Value
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Weight Decay	1e-5
Epochs	100
Cross Validation	5-Fold Stratified
Loss Function	Weighted Cross-Entropy
Device	CPU

Appendix B: Software and Libraries

The implementation was developed using Python and the following libraries:

Library	Purpose
PyTorch	Deep learning framework
NumPy	Numerical computation
Pandas	Data handling
Scikit-learn	Machine learning baselines
Matplotlib	Visualization
SHAP	Explainability analysis
XGBoost	Gradient boosting baseline

Appendix C: Generated Outputs

The framework generated the following outputs:

Output	Description
cv_metrics.png	Cross-validation performance
roc_pr_calibration.png	ROC, PR, and calibration curves
confusion_matrix.png	Classification confusion matrix
attention_gates.png	Modality gate distributions
permutation_importance.png	Modality importance analysis
shap_summary.png	SHAP feature explanations
baseline_model_comparison.csv	Classical ML comparison
best_model.pt	Saved trained model

Appendix D: Key Scientific Findings

The major findings of this study include:

1. Transcriptomics provided the strongest direct predictive signal.
2. Genomics contributed complementary contextual information despite low attention weights.
3. Classical machine learning models struggled with naive multimodal concatenation.
4. The proposed gated fusion architecture improved multimodal integration.
5. SHAP analysis revealed sparse but biologically meaningful feature contributions.
6. The framework achieved strong predictive performance with stable cross-validation behavior.

Appendix E: Potential Real-World Applications

The proposed framework may be applicable in:

- Precision oncology
- Biomarker discovery
- Disease subtype classification
- Personalized medicine
- Drug response prediction
- Clinical decision support systems
- Multi-omics disease diagnostics

Appendix F: Suggested Future Enhancements

Potential future improvements include:

1. Integration of graph neural networks.
2. Transformer-based multimodal fusion.
3. Pathway-aware representation learning.
4. Real quantum circuit simulation.
5. Federated biomedical learning.
6. Multi-task disease prediction.
7. Single-cell multi-omics analysis.
8. Temporal longitudinal patient modeling.

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